

Lecture-#4

Natural Language Processing

A8706

Introduction

Finding the Structure of Words and Documents



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Issues and Challenges

- **Irregularity:** word forms are not described by a prototypical linguistic model.
 - **Ambiguity:** word forms be understood in multiple ways out of the context of their discourse.
 - **Productivity:** is the inventory of words in a language finite, or is it unlimited?
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- **Irregularity:** Irregularity in language refers to situations where word forms do not follow the typical or regular patterns that might be expected based on linguistic rules or models.
 - Instead, irregular words have unique or idiosyncratic forms that don't conform to standard morphological or phonological rules. Here are some key points regarding irregularity in language:
 - Morphological Irregularity:** Irregularity often manifests in the way words are formed. For example, in English, the past tense of regular verbs is typically formed by adding "-ed" (e.g., "walk" becomes "walked"). However, irregular verbs like "go" have unique past tense forms ("went") that do not follow this pattern.
 - Phonological Irregularity:** Irregularity can also occur at the phonological level, affecting the pronunciation of words. For instance, the pronunciation of some irregularly spelled words in English may not align with their spelling (e.g., "colonel" is pronounced as "kernel").

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Ambiguity: Word structure can introduce ambiguity in language. The same word may have multiple meanings depending on its context and the affixes used.

For example, the word "**unhinged**" can mean "**mentally unstable**" or "**detached from its hinges**," illustrating how a single word can have different meanings based on its structure.

Lexical Ambiguity: Lexical ambiguity occurs when a single word has multiple meanings, and the context does not make it clear which meaning is intended. Examples include:

"Bank" can mean a financial institution or the side of a river.

"Bat" can refer to a flying mammal or a piece of sports equipment.

Syntactic Ambiguity: Syntactic ambiguity arises from the structure or arrangement of words within a sentence. Different interpretations of the sentence can result from alternative ways of parsing it. Examples include:

"I saw the man with the telescope." (Did I use the telescope to see the man, or did the man have the telescope?)

"The chicken is ready to eat." (Is the chicken ready to be eaten, or is it prepared to eat something else?)

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Semantic Ambiguity: Semantic ambiguity occurs when a sentence or phrase is vague or imprecise due to multiple possible interpretations of its meaning. Examples include:

"She saw the man with glasses." (Does "with glasses" refer to the man wearing glasses or the glasses themselves?)

"He's a little chicken." (Does "chicken" mean cowardly or a small bird?)

Pragmatic Ambiguity: Pragmatic ambiguity arises from the context or implied meaning of a statement. It can occur when a statement is technically clear but can be interpreted in different ways based on the speaker's intention or the situation. Examples include:

"Could you pass the salt?" (Is this a request for someone to pass the salt, or is it a polite way of saying, "I would like some salt, please"?)

"I am not saying he's a liar." (Is the speaker denying that the person is a liar, or are they implying something indirectly?)

Homonyms: Homonyms are words that have the same spelling or pronunciation but different meanings. This can lead to lexical ambiguity. Examples include:

"Bear" (referring to the animal) and "bear" (meaning to carry or endure).

"Tear" (referring to ripping something) and "tear" (referring to a drop of liquid from the eye).

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Productivity:

- ❖ Not all word formations are equally productive in a language. Some affixes are more versatile and can be applied to a wide range of words, while others have limited applicability.
- ❖ Understanding which affixes are productive and which are not is important for linguistic analysis and language learning.

Original word: **happy**

New word: **unhappy** (not happy)

Original word: **do**

New word: **undo** (reverse the action of doing)

Original word: **likely**

New word: **unlikely** (not likely)

Original word: **fair**

New word: **unfair** (not fair)

Original word: **able**

New word: **unable** (not able)

Original word: **pleasant**

New word: **unpleasant** (not pleasant)

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Morphological Models

- ❖ Morphological models in linguistics are theoretical frameworks or approaches that are used to describe and analyze the structure of words in a language.
- ❖ These models help linguists understand how words are formed, how they inflect, and how their internal components (morphemes) contribute to their meanings and functions.
- ❖ There are several morphological models, each with its own way of categorizing and analyzing linguistic elements. Here are some of the key morphological models:

• Dictionary Lookup

- Dictionary lookup is a practical application of morphological models in the context of language and linguistics.
- While dictionary lookup itself doesn't explicitly employ a specific morphological model, it relies on the underlying principles of morphology to provide users with information about words and their structures.

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Here's how dictionary lookup relates to morphological models :

- **Word Structure:** Morphological models describe how words are structured, including the identification of morphemes (the smallest meaningful units of language) and their combinations to form words.
- When you perform a dictionary lookup, the information you receive often includes details about a word's morphological structure. For example, you may learn about the word's root, affixes, and how they contribute to its meaning.
- **Inflection and Derivation:** Morphological models distinguish between inflectional and derivational morphemes.
- Inflectional morphemes convey grammatical information such as tense, number, and case, while derivational morphemes create new words or change word categories (e.g., noun to verb).
- Dictionaries often indicate whether a word has undergone inflectional changes and may provide derivational information about related words or word forms.

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Consider the English verb "**to run**":

Inflectional Forms:

Base form: **run**

Third-person singular: **runs**

Past tense: **ran**

Present participle: **running**

Past participle: **run**

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➤ **Word Forms:**

Nouns: Typically, nouns have singular and plural forms (e.g., cat vs. cats). Some languages also have other forms, like dual in Arabic.

Verbs: Verbs often have multiple forms to express different tenses, aspects, moods, or voices (e.g., walk vs. walked vs. walking).

Adjectives: Adjectives might change form to express degrees of comparison (e.g., happy, happier, happiest).

➤ **Lexical Ambiguity:** Morphological models help explain the different meanings that a word can have based on its morphological structure.

➤ When you look up a word in a dictionary, you may encounter multiple definitions or senses of the word, and understanding the morphological components can aid in disambiguating these meanings.

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Are Dictionary Lookup and Orthography the Same?

No, they are *not the same*, but they are *closely related* in morphological analysis.

Orthography-based Morphological Analysis

- **Orthography** deals with the **written form of words**.
- In morphological analysis, it means **breaking a word into parts (prefix, root, suffix) based on spelling patterns**.
- It **does not require meaning** — it's all about the **surface structure**.

Example:

"unhappily" → "un" + "happi" + "ly"

Even if "happi" is not a valid word, it's just based on **what remains after removing affixes**.

- **Focus:** Word structure by form
Used for: Rule-based stemmers, language teaching, quick heuristics

Dictionary Lookup in Morphology

- Here, you **check each word or morpheme against a known lexicon** (i.e., a dictionary).
- You **only recognize meaningful units** that exist in your dictionary.
- It verifies if the remaining **root or affix is a valid morpheme**.

Example:

"redo" → check if "re" is a known prefix and "do" is a valid root in dictionary.

Focus: Validity by meaning

Used for: Lemmatization, morphological analyzers (like spaCy, NLTK)

Finite-State Morphology

- Finite-State Morphology (FSM) is a computational linguistic model used for describing and generating complex word forms and morphological processes in natural languages. It employs finite-state automata and transducers to represent the morphological rules and processes. FSM is particularly useful for languages with relatively simple and regular morphological systems.
- Here's a simplified example of Finite-State Morphology in English, focusing on the pluralization of nouns:
- By finite-state morphological models, we mean those in which the specifications written by human programmers are directly compiled into finite-state transducers.
- The two most popular tools supporting this approach, XFST (Xerox Finite-State Tool) and Lex Tools.
- Finite-state transducers are computational devices extending the power of finite-state Automata.

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Example: Pluralization in English

In English, many nouns form their plurals by adding the suffix "-s" or "-es" to the singular form. Finite-State Morphology can be used to model this process.

Singular Nouns: The FSM model starts with a list of singular nouns, represented as follows:

"cat" -> "cat"

"dog" -> "dog"

"book" -> "book"

"house" -> "house"

Pluralization Rules: The FSM model includes rules for pluralization, which are typically defined as finite-state transducers. In this example, we have two rules:

Rule 1: Add "-s" to the singular form.

Rule 2: Add "-es" to the singular form if it ends with a consonant followed by "y" (e.g., "city" -> "cities").

Plural Nouns: The FSM applies the pluralization rules to the singular nouns to generate the plural forms:

Applying Rule 1 to "cat" -> "cats"

Applying Rule 1 to "dog" -> "dogs"

Applying Rule 1 to "book" -> "books"

Applying Rule 2 to "city" -> "cities"

Applying Rule 2 to "house" -> "houses"

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Input	Input Morphological parsed output
Cats	cat +N +PL
Cat	cat +N +SG
Cities	city +N +PL
Geese	goose +N +PL
Goose	goose +N +SG) or (goose +V)
Geese	goose +V +3SG
merging	merge +V +PRES-PART
Caught	(caught +V +PAST-PART) or (catch +V +PAST)

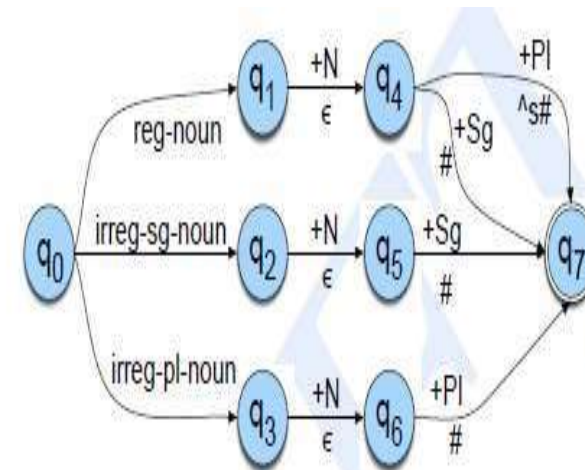


Figure 3.13 A schematic transducer for English nominal number inflection T_{num} . The symbols above each arc represent elements of the morphological parse in the lexical tape; the symbols below each arc represent the surface tape (or the intermediate tape, to be described later), using the morpheme-boundary symbol ϵ and word-boundary marker $\#$. The labels on the arcs leaving q_0 are schematic, and need to be expanded by individual words in the lexicon.

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- The FSM model captures the regularity of English pluralization by representing it as a set of finite-state rules and applying these rules to generate plural forms.
- While this example is highly simplified, FSM models can be extended to cover more complex morphological processes in various languages.
- FSM is commonly used in computational linguistics and natural language processing tasks, such as stemming, lemmatization, and inflection generation, where it can efficiently handle large sets of word forms and their associated morphological rules.

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Unification-Based Morphology

- Unification-Based Morphology (UBM) is a linguistic model that represents morphological processes using unification, a concept from formal logic and artificial intelligence.
- UBM focuses on the idea that morphological processes can be described as a series of feature unifications, where features of morphemes come together to form words.
- Morphological parsing P thus associates linear forms ϕ with alternatives of structured content ψ , cf.

I have seen two cats walks down the street

$$\mathcal{P} :: \phi \rightarrow \{\psi\}$$

$$\mathcal{P} :: form \rightarrow \{content\}$$

$$(1.2)$$

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- ❖ unification can be monotonic (i.e., information-preserving), or it can allow inheritance of default values and their overriding.
- ❖ In either case, information in a model can be efficiently shared and reused by means of inheritance hierarchies defined on the feature structure types.'
- ❖ Unification is the key operation by which feature structures can be merged into a more informative feature structure.
- ❖ Unification of feature structures can also fail, which means that the information in them is mutually incompatible.
- ❖ Morphological models of this kind are typically formulated as logic programs, and unification is used to solve the system of constraints imposed by the model.
- ❖ Advantages of this approach include better abstraction possibilities for developing a morphological grammar as well as elimination of redundant information from it.

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Unification:

- ❖ Unification is a process of merging two or more sets of features (attribute-value pairs) to form a single, consistent set. If the features are compatible, unification succeeds; if there is a conflict (e.g., a feature expecting both singular and plural values), unification fails.
- ❖ In morphology, unification is used to combine features from different morphemes or morphological rules to produce a complete word form or to analyze an existing word form.

Feature Structures:

- ❖ Feature structures are the primary data representations in unification-based systems. They consist of attributes (e.g., "number," "tense," "case") and their corresponding values (e.g., "singular," "past," "nominative").
- ❖ Feature structures can be complex, hierarchical, and recursive, allowing for the detailed representation of morphological, syntactic, and semantic information.

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Morphological Rules:

- ❖ In unification-based morphology, morphological rules are expressed as constraints on feature structures.
- ❖ These rules specify how morphemes combine and what features must agree (e.g., subject-verb agreement in number and person).
- ❖ For example, a rule might state that a noun's number feature must match the number feature of an affix (e.g., plural suffix "-s" in English).

Handling Agreement and Inflection:

- ❖ Unification is particularly powerful in handling agreement phenomena, where different parts of a sentence must share certain grammatical features. For instance, subject-verb agreement in English requires the subject and verb to agree in number and person.
- ❖ Unification-based morphology can also handle complex inflectional systems, where multiple features interact to produce the correct word form.

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Examples of Unification-Based Approaches:

- ❖ **Head-Driven Phrase Structure Grammar (HPSG):** A type of grammar that relies heavily on unification for both syntax and morphology. In HPSG, words are described by complex feature structures, and morphological rules are defined through unification constraints.
- ❖ **Lexical Functional Grammar (LFG):** Another formal grammar that uses unification to manage the interactions between different linguistic levels, including morphology.

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Consider the English word "walked":

- **Feature Structures:**

- The root "walk" might have features like:
 - ROOT: "walk"
 - POS: verb
- The past tense suffix "-ed" might have features like:
 - TENSE: past
 - POS: suffix

- **Unification:**

- The unification process would combine these feature structures to produce a complete feature structure for "walked":
 - ROOT: "walk"
 - POS: verb
 - TENSE: past

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Functional Morphology

- ❖ Functional morphology defines its models using principles of functional programming and type theory.
- ❖ It treats morphological operations and processes as pure mathematical functions and organizes the linguistic as well as abstract elements of a model into distinct types of values and type classes.
- ❖ Though functional morphology is not limited to modelling particular types of morphologies in human languages, it is especially useful for fusional morphologies.
- ❖ Linguistic notions like paradigms, rules and exceptions, grammatical categories and parameters, lexemes, morphemes, and morphs can be represented intuitively(without conscious reasoning; instinctively) and succinctly(in a brief and clearly expressed manner) in this approach.
- ❖ Functional morphology implementations are intended to be reused as programming libraries capable of handling the complete morphology of a language and to be incorporated into various kinds of applications.

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- ❖ Morphological parsing is just one usage of the system, the others being morphological generation, lexicon browsing, and so on.
- ❖ we can describe inflection I , derivation D , and lookup L as functions of these generic type

$$I :: \text{lexeme} \rightarrow \{\text{parameter}\} \rightarrow \{\text{form}\} \quad (1.3)$$

$$D :: \text{lexeme} \rightarrow \{\text{parameter}\} \rightarrow \{\text{lexeme}\} \quad (1.4)$$

$$L :: \text{content} \rightarrow \{\text{lexeme}\} \quad (1.5)$$

Many functional morphology implementations are embedded in a general-purpose programming language, which gives programmers more freedom with advanced programming techniques and allows them to develop full-featured, real-world applications for their models.

Function Morphology

- ❖ It is not a widely recognized term in linguistic theory, but it might refer to a conceptual or theoretical approach where morphological processes are understood and described in terms of functions, particularly in computational models of language. If this is what you're referring to, it can be interpreted in a couple of ways:

1) Morphological Functions:

In this interpretation, morphology is seen as serving specific functions within a language, such as creating new word forms, marking grammatical categories (like tense, number, or case), or generating derivational forms. Each morphological process (e.g., affixation, compounding) has a specific function in language structure and communication.

2) Morphology in Functional Grammar:

In functional grammar theories, such as those proposed by Simon Dik (Functional Grammar) or in more general functionalist approaches, morphology is analyzed in terms of its role in achieving communicative purposes. Here, morphological markers are seen as tools that help speakers express meanings, organize discourse, or maintain cohesion.

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3) Functional Programming Approaches to Morphology:

- ❖ In computational linguistics, the term might refer to using **functional programming** principles to model and implement morphological processes. In this context, "functions" are programming constructs that take inputs (e.g., root words, grammatical features) and produce outputs (e.g., inflected forms).

• **Morphological Functions** could be written as functions in a programming language that map morphemes and features to the appropriate word forms. For example:

```
pluralize("cat") -> "cats"  
past_tense("run") -> "ran"
```

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```
def past_tense(verb):  
    if verb == "run":  
        return "ran"  
    else:  
        return verb + "ed"  
  
def conjugate(verb, tense):  
    if tense == "past":  
        return past_tense(verb)  
    elif tense == "present":  
        return verb + "s" if verb.endswith("e") else verb + "es"  
    # Add more cases for other tenses as needed
```

